SE 216 – SOFTWARE PROJECT MANAGEMENT
PROJECT RISKS DOCUMENT

PROJECT NAME: **Happy Tails**

GROUP MEMBERS: **Semih Öztürk, Mirza Kerem Kılıç, Melike Gülmine Kırpat, Özge Öztürk, Hüseyin Günçiçek, Arda Özkaranfil, İdil Buse Zengin**

LIKELIHOOD RANK	RISK DESCRIPTION
1	**Requirements Volatility:** Shelter management rules or adoption workflows may change during development. Specifically, moderation rules for the "Story Tail" community section may shift as the platform grows.
2	**AI Content Control Accuracy:** The OpenAI-based content control for user stories might fail to filter inappropriate content or incorrectly flag valid posts, impacting user trust in the "Story Tail" feature.
3	**Third-Party API Dependency:** External APIs might update their version, change their pricing model, or suffer an outage that is completely out of our control.
4	**Logic Conflicts in Software Structure:** Complex business logic across modules such as AI matching scoring, donation allocation, inventory state transitions, and concurrent user actions may produce unexpected outcomes. These conflicts can be difficult to detect early and may require significant debugging effort during integration.
5	**Race Condition:** Two different processes are trying to modify the same data at the same time.
6	**Testing Environment Discrepancies:** The product may be difficult to test effectively if the deployment platform is not fully acquired or configured to match the final production environment.
7	**Hardware & Connectivity Issues:** Potential difficulties in maintaining a stable connection between the physical shelter cameras and the application's backend architecture.
8	**System Availability & Architecture Bottlenecks:** The integration of live monitoring and AI filtering might cause performance issues, making it difficult to maintain the high availability required for a 24/7 shelter app.
9	**Version and Dependency Conflicts:** The risk that conflicting library versions or breaking updates in the software stack (such as Flutter packages or Firebase SDKs) stall development by forcing teams to resolve technical overlaps instead of building new features.
10	**Data Security & Encryption Complexity:** Challenges in correctly implementing SHA-256 encryption across all user data while maintaining system performance and availability.

IMPACT RANK	RISK DESCRIPTION
1	**AI Content Control Accuracy:** Errors in the OpenAI-based moderation system could lead to the public sharing of inappropriate content, damaging the project's reputation and user safety.
2	**Data Security & Encryption Complexity:** Failure to correctly implement SHA-256 encryption could lead to catastrophic data breaches and legal non-compliance.
3	**System Availability & Architecture Bottlenecks:** Performance issues in the core architecture could prevent the app from maintaining 24/7 availability, rendering the entire platform unusable.
4	**Third-Party API Dependency:** External APIs might update their version, change their pricing model, or suffer an outage that is completely out of our control.
5	**Requirements Volatility:** While shifts in shelter rules or workflows cause delays, they usually require process adjustments rather than causing total system failure.
6	**Hardware & Connectivity Issues:** Physical connectivity problems are often localized or temporary and can be fixed without redesigning the software architecture.
7	**Logic Conflicts in Software Structure:** Complex business logic across modules such as AI matching scoring, donation allocation, inventory state transitions, and concurrent user actions may produce unexpected outcomes. These conflicts can be difficult to detect early and may require significant debugging effort during integration.
8	**Version and Dependency Conflicts:** The risk that conflicting library versions or breaking updates in the software stack (such as Flutter packages or Firebase SDKs) stall development by forcing teams to resolve technical overlaps instead of building new features.
9	**Race Condition:** Two different processes are trying to modify the same data at the same time.
10	**Testing Environment Discrepancies:** If the deployment platform is misconfigured compared to production, the product may fail completely at the point of launch.

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LIKELIHOOD RANK	IMPACT RANK	COMBINED RANK	RISK DESCRIPTION
2	1	3	**AI Content Control Accuracy:** Errors in the OpenAI-based moderation system could lead to the public sharing of inappropriate content, damaging the project's reputation and user safety.
1	5	6	**Requirements Volatility:** While shifts in shelter rules or workflows cause delays, they usually require process adjustments rather than causing total system failure.
3	4	7	**Third-Party API Dependency:** External APIs might update their version, change their pricing model, or suffer an outage that is completely out of our control.
4	7	11	**Logic Conflicts in Software Structure:** Complex business logic across modules such as AI matching scoring, donation allocation, inventory state transitions, and concurrent user actions may produce unexpected outcomes. These conflicts can be difficult to detect early and may require significant debugging effort during integration.
8	3	11	**System Availability & Architecture Bottlenecks:** Performance issues in the core architecture could prevent the app from maintaining 24/7 availability, rendering the entire platform unusable.
10	2	12	**Data Security & Encryption Complexity:** Failure to correctly implement SHA-256 encryption could lead to catastrophic data breaches and legal non-compliance.
7	6	13	**Hardware & Connectivity Issues:** Physical connectivity problems are often localized or temporary and can be fixed without redesigning the software architecture.
5	9	14	**Race Condition:** Two different processes are trying to modify the same data at the same time.
6	10	16	**Testing Environment Discrepancies:** If the deployment platform is misconfigured compared to

LIKELIHOOD RANK	IMPACT RANK	COMBINED RANK	RISK DESCRIPTION
			production, the product may fail completely at the point of launch.
9	8	17	**Version and Dependency Conflicts:** The risk that conflicting library versions or breaking updates in the software stack (such as Flutter packages or Firebase SDKs) stall development by forcing teams to resolve technical overlaps instead of building new features.